

VRUSHAL VIJAY DALVI

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PROFESSIONAL SUMMARY

Pharmaceutical scientist with graduate research in scalable production, formulation and biophysical characterization of protein-loaded lipid nanoparticles (LNPs) for parenteral delivery. Hands-on with LNP formulation and lipid biochemistry, encapsulation efficiency, particle size and hydrodynamic analysis, and stability assessment of colloidal systems using SEC, HPLC, DLS, circular dichroism, intrinsic fluorescence, UV-Vis, and FTIR spectroscopy. Skilled in analytical method development and literature-based method adaptation, mammalian cell culture under aseptic technique (passaging, seeding, banking), ELISA, and MicroBCA protein quantification. Experienced in cGMP workflows, SOP review, and Design of Experiments (DoE) for formulation development.

EDUCATION

State University of New York (SUNY) at Buffalo, NY

Aug 2024-Jul 2026

MS Pharmaceutical Sciences (*GPA 3.40/4*)

Courses: Intermed. Pharmacokinetics, Biologics, Pharmacogenomics, Pharmacometrics Software, Biometry

Institute of Chemical Technology (Formerly UDCT), Mumbai

Dec 2022

M. Tech Speciality Chemicals Technology (*CGPA 9.55/10*)

University of Mumbai

May 2019

Bachelor of Pharmacy (*CGPA 7.32/10*)

PROFESSIONAL EXPERIENCE

Eagle Wings Enterprises LLP, India | Junior Perfumer

Nov 2021-Jun 2024

- Developed market-specific fragrance formulations for cross-regional portfolios (India, Middle East, Sweden, Sri Lanka, Bangladesh), directly contributing to >30% year-over-year revenue growth.
- Built and curated a reference library of 200+ aroma chemicals and integrated structured datasets to enable systematic olfactory matching and formulation screening.
- Conducted sensory evaluation panels to optimize fragrance profiles across product categories, informing formulation iterations based on structured feedback.

FDC Limited, India | In-plant trainee

Apr 2018-Jun 2018

- Supported cGMP manufacturing operations for pharmaceutical products designated for US, UK, and Indian markets; reviewed SOPs to understand region-specific regulatory requirements thereby gaining hands-on exposure to end-to-end line operations and multi-region compliance workflows.
- Assisted in aseptic manufacturing of ophthalmic solutions using Blow-Fill-Seal (BFS) technology under controlled cleanroom conditions.
- Supported QC/QA analytical testing using GC, HPLC, and Karl Fischer Coulometry, contributing to a 5% reduction in quality variances and improved batch record accuracy.

RESEARCH PROJECTS

1. Scalable Production & Bio-characterization of Lysophosphatidylserine Nanoparticles Containing Ovalbumin for Immune Tolerance (Thesis)

Jan 2025-May 2026

- Designed and evaluated 18:1 LysoPS-DMPC lipid nanoparticles loaded with Ovalbumin as model protein; developed and optimized an extrusion-free thin-film hydration process to address a scalability gap for large-scale manufacturing.
- Built a multi-technique biophysical characterization workflow to define LNP quality attributes like hydrodynamic size by DLS, surface phosphatidylserine exposure via PSvue™ probe fluorescence and membrane fluidity by DPH anisotropy.

- Quantified protein association efficiency by Micro BCA and validated native protein conformation post-association using circular dichroism and intrinsic fluorescence spectroscopy; benchmarked extrusion-free outputs against extruded controls to confirm preservation of critical biophysical attributes.
- Assessed in-vitro tolerogenic potential through dendritic cell assays using multi-color flow cytometry under aseptic cell culture conditions, profiling surface marker expression to compare immunomodulatory activity across formulation methods.

2. Optimization of Enzymatic Synthesis of Perfumery Esters Using RSM (Thesis) Aug 2020-Nov 2021

- Designed and executed multi-parameter enzymatic assays using lipases from *Aspergillus oryzae* and *Candida antarctica*, systematically screening reaction conditions across temperature, substrate ratio, and enzyme loading variables.
- Applied Response Surface Methodology (RSM) to model parameter interactions, reducing reaction time by 25% and enzyme load by 15% while achieving yields >92%.
- Ran solvent-free enzymatic reactions to reduce hazardous wastes and maintain industrial viability.
- Verified product purity via GC analysis and compiled a reproducible, scalable protocol.

3. Studies on Green Method of Synthesis of Schiff's Base Using Natural Acid Catalyst Sept 2018-Feb 2019

- Implemented green synthesis of hydrazone derivative using natural fruit juice catalysts.
- Achieved ~30% faster reaction rates versus conventional reaction conditions.
- Optimized reaction yield by ~15% using citric acid from lemon juice, identifying it as the most effective catalyst with a reaction time ~25% faster compared to other acids.
- Validated hydrazone purity (95%) using IR spectroscopy, showcasing the viability of natural acid catalysts.

TECHNICAL SKILLS

- **Cell-Based & Biochemical Assays:** Mammalian Cell Culture (aseptic technique, passaging, seeding), Cell Viability Assays, ELISA, Micro BCA Protein Quantification, Imaging Cytometer, Flow Cytometry.
- **Biophysical & Assays:** LNP formulation, Particle sizing, PSvue™, DPH Anisotropy, GC-MS, HPLC, FTIR, UV-Vis Spectrophotometer, DLS, Spectrofluorometer, TEM, Circular Dichroism, SEC.
- **Modelling & Simulations:** Phoenix™WinNonlin, ADAPT 5, SimBiology, GastroPlus®, Monolix™, Symcyp®.
- **Data & Productivity:** Microsoft Suite, R Programming, GraphPad Prism®, ChemDraw, RSM.

TECHNICAL PROJECT HIGHLIGHTS

- **ADAPT 5 | Target Mediated Drug Disposition (TMDD) & Michaelis-Menten (MM) Approximations 2025**
Modelled and simulated TMDD rapid binding and Michaelis-Menten models for different doses, varied KD and compared the outputs for drug concentration, drug-receptor concentration, free receptor concentration and receptor occupancy.
- **SimBiology | Antibacterial PK of Doripenem 2025**
Fit a basic PK model and built a bacterial growth/kill model with K_{max} and $KC50$ for a dosing of 12 hours, ran a 0-500 mg parameter scan and generated an analysis report.
- **Phoenix WinNonlin | Remifentanyl PK 2024**
Performed NCA and built learnt to build a one compartment and two compartment infusion model, compared the fit and captured final parameter estimates.

CERTIFICATIONS & ADVANCED COURSEWORK

- Advanced Topics in PK/PD Modeling & Analysis with ADAPT 5 (2025)
- SimBiology (2025)
- Modelling & Simulations using Monolix Suite™ (2025)
- PKPD Modeling Workshop Using Phoenix (2024)
- Safety & Risk Management (2020)
- Poster Presentation: "Green Method of Synthesis of Schiff's Base Using Natural Acid Catalysts" (2019)
- Poster Presentation: "Novel Drug Delivery Systems" (2017)